MIL-A-82151B(MC)
12 October 1989
SUPERSEDING
MIL-A-82151A(MC)
15 September 1969

MILITARY SPECIFICATION

ANTENNA: AS-2236, 30 TO 76 MEGAHERTZ

This specification is approved for use by the U. S. Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type antenna, man or vehicular transportable.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-T-152 - Treatment, Moisture and Fungus-Resistant, of Communications, Electronic and Associated Electrical Equipment.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding General (PSE-C), Marine Corps Research, Development and Acquisition Command, Washington, D.C. 20380-0001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5985

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MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment
MIL-E-16400	- Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore: General Specification for.
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of.
MIL-E-21981	- Electronics Equipment, Nomenclature, Serial Numbers and Identification Plates: Requirements of.
MIL-P-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.

STANDARDS

MILITARY

MIL-STD-810	- Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

261D773	- Antenna, AS-2236/GRC
MI-06728A-35/2A	- Reinforcing the Antenna Subassembly, Antenna, AS-2236/GRC.

(Copies of Government documents, drawings, and publications required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 FIRST ARTICLE. When specified (see 4.2.1), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.1.

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3.2 Parts and materials. In the selection of parts and materials, fulfillment of major objectives shall be the prime consideration. In all cases, general material requirements as to flammability, fungus-resistance and nontoxicity as required by MIL-E-16400 shall be met.

3.2.1 Mechanical parts. Where mechanical parts whose function is to rotate, cam or oscillate against others are employed, and the best choice of materials includes other than those listed in MIL-E-16400, use of the other materials may take precedence. The contractor shall submit evidence that improved maximum life will be achieved by use of lubricating agents, protective paints, sprays or other methods.

3.2.2 Nonstandard parts. When it has been determined that requirements cannot be met by use of parts and materials as specified in MIL-E-16400, immediate action shall be taken by the contractor to inform the contracting officer by submission of detailed data, as required by MIL-E-16400, requesting approval.

3.2.3 Fungus treatment. The equipment shall be fungus proofed by selection of parts and materials which are nonnutrient for fungus, or the parts and materials used shall be treated in accordance with MIL-T-152 prior to use in the equipment.

3.3 Design and construction. The Antenna, AS-2236, PN 85226/261D773 shall consist of a log periodic directional antenna, a vertically erected mast, a tripod assembly and the necessary ancillary equipment to provide a complete antenna. Also, during the construction of antenna subassembly, MI-06728A-35/2A shall be incorporated. It shall be capable of providing directional antenna performance. The antenna shall be capable of handling all radio frequency power levels up to at least 65 watts, continuous wave, over the 30 to 76 megahertz (MHz) spectrum. The antenna shall consist of the following:

Item No.	Item	Qty.
1	Tripod Assembly	1
2	Mast Assembly	1
3	Antenna Assembly	1
4	Guy Assembly	1
5	Guy Anchor Set	1
6	Tripod Anchor Set	1
7	Carrying Cases Set	1
8	Radio Frequency (RF) Cable Assembly	1
9	Earth Anchor Setting Implement	1
10	Polarizer	1
11	Tripod Brace	1

3.3.1 Tripod assembly. The tripod assembly shall be a lightweight unit consisting of a mast launching assembly and tripod base. The launching assembly shall provide guides for the mast section

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during erection of the antenna. A vertical restraint device shall be provided to support the antenna mast assembly in the extended position. The launching section shall be equipped with a clamping mechanism for maintaining azimuth orientation of the antenna, while providing for intermediate vertical restraint of the antenna. The lauching section shall be capable of supporting the antenna at any incremental height up to its fully extended position. The tripod assembly shall be sufficiently rigid to support the antenna at its fully extended height of 35 feet in horizontal wind conditions of 20 knots, without guying. No damage shall be sustained by any component of the antenna during wind conditions up to 60 knots, horizontal, at its fully extended height, when the guy cable assembly is employed.

3.3.2 Tripod legs. The tripod shall consist of three folding, telescoping tripod legs. A minimum of 12 inches of telescopic adjustment shall be provided in each leg to permit leveling the base on uneven terrain. Locking of each leg at its desired height shall be accomplished with the use of a hand actuated device. No tools shall be required. The legs shall be provided with drilled, swiveled feet to allow staking the tripod into the earth. The foot pads of the tripod shall measure between 8 and 9 inches square. The tripod shall be capable of erection, adjustment and disassembly for transport without the use of tools.

3.3.3 Tripod brace. A means shall be provided for connecting the lower portion of the tripod legs to each other to form a triangular truss. The device is intended to prevent the tripod footpads from sliding with respect to each other when soil conditions reduce the effectiveness of footpads stakes. The device is to consist of flexible steel cables with a tension adjusting device.

3.3.4 Mast assembly. The mast assembly shall consist of eight readily detachable, mechanically interchangeable sections of such dimensions as to adequately support the antenna at its fully extended height of 35 feet.

3.3.4.1 Lower mast sections. The six lower mast sections shall be provided with keyed couplings so as to provide positive azimuth orientation of the antenna.

3.3.4.2 Upper mast sections. The two upper mast sections shall be made of a dielectric material so as to prevent distortion of the antenna pattern when the antenna is placed in either the horizontal or vertical plane. They shall be provided with keyed couplings so as to maintain positive azimuth orientation of the antenna. The dielectric mast sections shall provide accommodation for the attachment of the guy cable assembly.

3.3.5 Guy assembly. The guy assembly shall consist of three dielectric rope guy lines, tensioning devices and mast mounting hardware sufficient to maintain the antenna upright in a 60 knot wind when in its fully extended position.

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3.3.6 Earth anchors

3.3.6.1 Guy anchor set. Three auger-type earth anchors shall be provided to fasten the guy cables. They shall be at least 18 inches in length and have a helix plate 3 inches in diameter. A hook shall be provided in the top of the shaft for attachment of the guy cables. The anger shall be capable of resisting a 300 pound force when installed with the antenna in loose, sandy soil.

3.3.6.2 Tripod anchor set. Six spike-shaped stakes shall be provided for holding the tripod footpads in place. The stakes shall fit through the holes in the footpads and have a head or other feature to capture the footpad securely in place and shall be driven by using a 2-1/2 pound hammer. These stakes shall be approved by the procuring activity.

3.3.7 Carrying case set. Three carrying cases shall be provided for the purpose of transporting all components of the antenna. The carrying cases shall be provided with carrying handles and shoulder straps. An accessory bag, or suitable substitute, which attaches to, or stows within, one the carrying cases shall be provided.

3.3.8 Radio frequency cable assembly. The RF cable assembly shall be 125 feet in length, constructed of RG-8A/U Coaxial Cable, and terminated with a type BNC male connector on each end. A clamp or similar device shall be installed at the upper cable end of the mast to eliminate strain on the connector caused by the weight of the dangling cable.

3.3.9 Polarizer. A device shall be provided which permits the polarization of the antenna to be changed without diassembling the mast or removing any guy lines. The device shall be mounted atop the mast just below the antenna and may be operated by ropes hanging down to the ground. The design shall insure that polarization position cannot be changed accidentally by wind factors.

3.3.10 Earth anchor setting implement. A 2-1/2 pound hammer shall be provided to drive the spike-shape stakes into the ground.

3.4 Reliability. Reliability shall receive major emphasis during the design, fabrication and construction of antennas, masts and ancillary equipment. This emphasis shall be reflected in the assembly processes and the determination of operational techniques. The first articles and selected sample equipments shall be subjected to complete assembly, erection and disassembly 100 times immediately prior to the mechanical and electrical tests required by Section 4 of this specification. Mechanical wear shall be such that mechanical or functional performance is not affected. Abrasion of painted surfaces is allowable; grooving or cutting or cammed, joined or sliding metal surfaces, or cracking or breaking of any portion of the antenna shall not be allowable. Electrical performance of the

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antenna shall not be affected in any way. Earth anchor bending, breaking or deformation is allowable.

3.5 Performance. The antenna shall be a lightweight, tactical type, high performance antenna covering the frequency range of 30 to 76 MHz. The antenna shall provide the following:

(a) Transmission and reception of horizontally or vertically polarized RF signals without tuning of the antenna, transmission line or matching device.

(b) Capable of being erected by two men in less than 15 minutes.

(c) Simple azimuth orientation.

(d) Capable of operation in winds of up to 33 knots without guying, or 60 knots with guying provided as part of this antenna at its fully erected height of 35 feet.

(e) A minimum of 4.5 decibels (dB) isotropic gain.

(f) A minimum front-to-back ratio of 8 dB.

(g) A maximum Voltage Standing Wave Ratio (VSWR) of 3:1.

(h) A maximum total weight of not more than 115 pounds and packaged so that it may be easily transported by three men.

3.5.1 Electrical performance. The electrical characteristics of the antenna shall be in accordance with this specification. All characteristics stated shall be required when the antenna is erected at its fully extended height of 35 feet above the terrain, in either the horizontal or vertical plane.

3.5.1.1 Forward gain. The antenna shall exhibit not less than 4.5 dB forward isotropic gain over the frequency range of 30 to 35 MHz and not less than 5.0 dB forward isotropic gain over the frequency range of 35 to 76 MHz.

3.5.1.2 Characteristic impedance. The antenna shall present a nominal load of 50 ohms, measured at the input connector, over the frequency range of 30 to 76 MHz.

3.5.1.3 Front-to-back ratio. The antenna shall provide a front-to-back ratio of not less than 7.5 dB over the frequency range of 30 to 76 MHz.

3.5.1.4 Voltage standing wave ratio. The antenna shall present a VSWR, measured at the input connector of the RF cable assembly, of not greater than 3:1 over the frequency range of 30 to 76 MHz.

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3.6 Dimensions. Dimensions of major components shall be as follows:

(a) Tripod assembly, when collapsed and packaged for trans- transport, external dimensions shall not exceed 15 inches by 48 inches maximum.

(b) Mast assembly, when collapsed and packaged for transport, external dimensions shall not exceed 12 inches by 48 inches maximum.

(c) Antenna assembly, when collapsed and packaged for transport external dimensions shall not exceed 10 inches by 48 inches maximum.

3.7 Weight. The maximum weights of the antenna components shall be as follows:

Item No.	Item	Max. Weight, Pounds
1	Tripod Assembly	21
2	Mast Assembly	27
3	Antenna Assembly	15
4	Guy Cable Assembly	3-1/2
5	Guy Anchor, Set	4-1/2
6	Tripod Anchor, Set	3
7	Carrying Cases, Set	16
8	Radio Frequency Cable Assembly	13-1/2
9	Earth Anchor Setting Implement	3
10	Polarizer	4
11	Tripod Brace	2-1/2

3.8 Finish. Metal surfaces on the tripod, mast and antenna shall be painted with Chemical Agent Resistant Coating (CARC), MIL-C-46168. Color shall match Marine Corps Green 383, No. 34094 (see 6.3). Surfaces subject to sliding wear, such as metal mast sections, tripod footsteps and earth anchors may, alternatively, be anodized Marine Corps Green 383 color in accordance with MIL-A-8625, Type III.

3.9 Identification marking. Subassemblies shall be identified with the following information rubber stamped or silk screened on a fixed surface, if available, or on a permanent tag affixed in such a manner as not to impair operation of the assembly. The marking shall not be easily removable.

(a) Nomenclature

(b) Approved Item Name

(c) Additional information required; Federal Stock No., Drawing No., manufacturer's name, and serial number.

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3.9.1 Identification plates. Identification plates shall be in accordance with MIL-P-15024 and shall be of aluminum alloy where practicable. Drawings for identification plates shall be submitted for approval 60 days prior to commencement of first article tests. Nomenclature shall be requested in accordance with MIL-E-21981 and shall be submitted 90 days prior to commencement of first article testing.

3.10 Workmanship. Antennas covered by this specification shall be manufactured and finished in accordance with MIL-E-16400.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsible for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein classified as follows:

a. First article inspection (see 4.2.1).

b. Production (see 4.2.2).

4.2.1 First article. Four antennas shall be used to complete the tests indicated for first article in Table I. First article testing will be witnessed by a representative of the contracting officer.

4.2.2 Production. All antennas shall be subjected to the tests indicated for production in Table I.

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TABLE I - TESTS

Performance	Reqt. Para.	Vibration	Shock	Temperature Shock	Humidity	Salt Spray	Rain	Wind Loading	Standard Conditions
Assembly erection and disassembly	3.4								FA S
Forward gain	3.5.1.1			FA S	FA S	FA S	FA S	FA S	FA S
Characteristic impedance	3.5.1.2				FA S	FA S	FA S		FA S
Front to back ratio	3.5.1.3				FA S	FA S	FA S		FA S
Voltage standing wave ratio	3.5.1.4				FA S	FA S	FA S		FA S P
Erection time	3.5								FA S
Mechanical and visual inspection	various	FA S	FA S	FA S	FA S	FA S		FA S	FA S P
Items to be tested									
Tripod assembly		X	X		X	X	X	X	X
Mast assembly		X	X		X	X	X	X	X
Antenna assembly		X	X	X	X	X	X	X	X
Guy cable assembly		X	X			X		X	X
Earth anchors		X	X					X	X
Carrying cases		X	X						X
RF cable assembly		X	X	X	X	X	X	X	X
Earth anchor setting implement		X	X						X

Note: P= Production FA= First Article S= Sample

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4.3 Tests

4.3.1 Vibration. Each antenna to be subjected to the vibration test shall be disassembled, stowed in its carrying cases and rigidly mounted to the vibration table in the horizontal plane. It shall be subjected to vibration in the frequency range of 10 to 55 Hertz (Hz) at 5 gravity (g), or 0.06 inches total excursion, whichever is greater. Frequency change shall be at a rate of less than one Hz. Duration of the test shall be one-half hour. There shall be no destructive rigid body resonance exhibited below 55 Hz.

4.3.2 Shock. Each antenna, with components, to be subjected to the shock test shall be disassembled and stowed in the carrying cases. It shall withstand a 3-hour test on the LAB package tester. For this test, the carrying cases shall be rigidly mounted to the adapter plate. The machine shafts shall be adjusted in speed and operated in phase to produce average peak accelerations of 10 g's with occasional peaks of 20 g's.

4.3.3 Temperature shock. The equipment shall be tested in accordance with Method 503.2, Procedure I of MIL-STD-810 except that the high temperature shall be +65°C.

4.3.4 Humidity. Antennas shall be tested in accordance with Method 507.2, Procedure I of MIL-STD-810 except the maximum temperature shall be +65°C. The test will be concluded at the end of three cycles. Any external moisture present may be removed by wiping.

4.3.5 Salt Fog. Place equipment in test chamber in disassembled condition and perform this test in accordance with Method 509.2, Procedure I of MIL-STD-810. After removal from the test chamber, electrical performance of the antenna shall not be degraded more than 15 percent. When salt accumulation is removed by washing or hosing with fresh water, the antenna electrical performance must meet the standards prescribed in 3.5.1.

4.3.6 Rain. Place the antenna in its normal operational position and perform this test in accordance with Method 506.2, Procedure I of MIL-STD-810. Allow simulated rainfall to fall on the antenna at a minimum rate of 2 (in/hr) for 1 hour. Immediately after completion of the simulated rainfall, subject the antenna group to all electrical test requirements of 3.5.1. Performance shall not be degraded below the prescribed standards. All electrical tests must be completed in less than 4 hours after soaking. Resoaking will be allowed.

4.3.7 Wind. The antenna system shall be subjected to the loading placed on it by a 60 knot wind blowing horizontally, for a period of 1 hour. At the end of this test, the antenna system shall show no evidence of electrical or mechanical degradation. Compliance with this procedure for the antenna mast may be demonstrated by appropriate calculation and simulation.

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4.4 Approval. Approval of the first articles shall be by the contracting officer upon satisfactory completion of all the tests.

4.5 Acceptance. With exception of the first articles, acceptance of all items shall be by the cognizant Government inspector.

5. PREPARATION FOR DELIVERY

5.1 General. All major units and parts of equipments shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555.

6. NOTES

6.1 Intended use. This antenna is to be employed as a directional, lightweight, man-transportable antenna for use with tactical radio equipment operating in the frequency spectrum of 30 to 76 MHz.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Whether first article is required (see 3.1).

(c) Level of preservation, packaging and packing (see 5.1).

6.3 Color. Color chips for Marine Corps Green 383, No. 34094 may be obtained from the Commandant of the Marine Corps (CSY-10), Headquarters U. S. Marine Corps, Washington, D.C. 20380-0001.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.2. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.4.1 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue doe to the extensiveness of the changes.

Preparing activity:
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SPECIFIC COMMENTS

1. Answers to specific comments outlined in the previous correspondence dated 15 May 1989 are as follows:

a. Metric System Conversion for this specification is not required. The technical data package for the antenna has not been converted and is in "inch-pounds" and the specification should correspond with the drawings.

b. Per your request, paragraph 4.2.3, 4.2.3.1 and 4.2.3.2 were deleted from the specification.

c. Under paragraph 4.2.3.1, the meaning of the phrase "one additional antenna for each 100 pounds" contained a typographical error. The word "pounds" should have read "products". This paragraph as stated in paragraph 1b has been deleted so no further action is required.

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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